TMUA PRACTICE TEST SERIES

TMUA Topic Test 5

Mixed-Topic Paper 1

- 20 multiple-choice questions
- Recommended time: 75 minutes
- All-topic challenge set
- Rigorous corrected solutions

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

TMUA Topic Test 5 — Mixed-Topic Set

This solution book matches the final TMUA Topic Test 5 question set and provides fully worked, non-calculator solutions with embedded supporting figures where relevant.

Answer key

1 **B**	2 **A**	3 **D**	4 **D**	5 **D**	6 **D**	7 **D**	8 **A**	9 **C**	10 **D**
11 **H**	12 **E**	13 **C**	14 **C**	15 **D**	16 **D**	17 **D**	18 **D**	19 **C**	20 **B**

Algebra and Structure

Polynomial roots, binomial coefficients, tangents, coordinate geometry and indices.

Q1

Polynomial roots

Answer **B**

Question 1

How many positive real roots does

$$f(x) = x^4 - 8x^3 + 22x^2 - 24x$$

have?

- **A** 0
- **B** 1
- **C** 2
- **D** 3
- **E** 4

WORKED SOLUTION

Factor out the obvious root:

$$f(x) = x\left(x^3 - 8x^2 + 22x - 24\right).$$

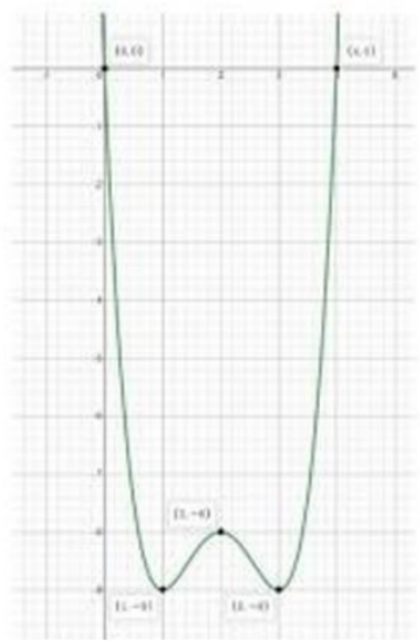
Since the cubic vanishes at $x = 4$,

$$x^3 - 8x^2 + 22x - 24 = (x-4)(x^2 - 4x + 6).$$

The quadratic has discriminant

$$(-4)^2 - 4(1)(6) = -8 < 0,$$

so it has no real roots. The real roots are therefore 0 and 4, of which only 4 is positive. Hence the answer is $\boxed{\text{B}}$.



The graph confirms the real roots x=0 and x=4; only x=4 is positive.

Q2

Binomial coefficients

Answer **A**

Question 2

In the expansion of $(2x+b)^9$, the coefficient of x^5 is equal to the coefficient of x^3 in $(3b+x)^6$.

Given that $b > 0$, find b.

A $\dfrac{15}{112}$

B $\dfrac{112}{15}$

C $\dfrac{15}{56}$

D $\dfrac{56}{15}$

E $\dfrac{5}{42}$

WORKED SOLUTION

The coefficient of x^5 in $(2x+b)^9$ is

$$\binom{9}{5}2^5b^4 = 126 \cdot 32b^4 = 4032b^4.$$

The coefficient of x^3 in $(3b+x)^6$ is

$$\binom{6}{3}(3b)^3 = 20 \cdot 27b^3 = 540b^3.$$

Equating and using $b > 0$,

$$4032b^4 = 540b^3 \implies b = \frac{540}{4032} = \frac{15}{112}.$$

The answer is $\boxed{\text{A}}$.

Q3

Tangents and inequalities

Answer **D**

Question 3

Consider the tangent to the curve $y = x^2 + bx$ at $x = 2$.

For which values of b is the x-intercept of the tangent greater than 4?

- **A** $-3 < b < 3$
- **B** $3 < b < 4$
- **C** $b > 3$
- **D** $-4 < b < -3$
- **E** $b > -3$

WORKED SOLUTION

At $x = 2$, the point is $(2, 4 + 2b)$, and the gradient is $4 + b$. Hence the tangent is

$$y - (4 + 2b) = (4 + b)(x - 2).$$

Setting $y = 0$, its x-intercept is

$$x = 2 - \frac{4 + 2b}{4 + b} = \frac{4}{4 + b}.$$

We require $\frac{4}{4+b} > 4$. The denominator must first be positive, and then

$$\frac{1}{4 + b} > 1 \quad \Longrightarrow \quad 0 < 4 + b < 1.$$

Thus $-4 < b < -3$, so the answer is $\boxed{\text{D}}$.

Q4

Coordinate geometry

Answer **D**

Question 4

The line $y = mx + 1$ is tangent to the circle

$$(x-2)^2 + (y+1)^2 = 5.$$

There are two possible values of m. What is their sum?

- **A** -8
- **B** -1
- **C** 1
- **D** 8
- **E** 16

WORKED SOLUTION

Write the line as $mx - y + 1 = 0$. The circle has centre $(2, -1)$ and radius $\sqrt{5}$. Tangency requires the distance from the centre to the line to equal the radius:

$$\frac{|2m+2|}{\sqrt{m^2+1}} = \sqrt{5}.$$

Squaring gives

$$4(m+1)^2 = 5(m^2+1) \quad \Longrightarrow \quad m^2 - 8m + 1 = 0.$$

By Vieta's formula, the sum of the two possible gradients is 8. The answer is $\boxed{\text{D}}$.

Q5

Indices and series

Answer **D**

Question 5

Suppose

$$5^{\,2+5+8+\cdots+(3x-1)} = 0.04^{-50},$$

where x is a positive integer. Find x.

- **A** 5
- **B** 6
- **C** 7
- **D** 8
- **E** 9

WORKED SOLUTION

The exponent on the left is the sum of an arithmetic progression with x terms, first term 2, and last term $3x-1$:

$$S = \frac{x}{2}\bigl(2+3x-1\bigr) = \frac{x(3x+1)}{2}.$$

Also, $0.04 = \frac{1}{25} = 5^{-2}$, so

$$0.04^{-50} = 5^{100}.$$

Therefore

$$\frac{x(3x+1)}{2} = 100 \quad\Longrightarrow\quad 3x^2 + x - 200 = 0 \quad\Longrightarrow\quad (3x+25)(x-8) = 0.$$

The positive integer solution is $x = 8$. The answer is $\boxed{\text{D}}$.

Probability, Sequences and Graphs

Conditional probability, trigonometric symmetry, recurrences and root counting.

Q6

Conditional probability

Answer **D**

Question 6

Three distinct cards are chosen uniformly at random from cards numbered $1, 2, \ldots, 7$.

Given that the product of the three chosen numbers is even, what is the probability that their sum is even?

A $\frac{12}{31}$

B $\frac{15}{31}$

C $\frac{18}{31}$

D $\frac{19}{31}$

E $\frac{20}{31}$

WORKED SOLUTION

There are $\binom{7}{3} = 35$ triples. The product is odd only when all three cards are odd; there are $\binom{4}{3} = 4$ such triples. Hence the condition leaves

$$35 - 4 = 31$$

possible triples.

A sum of three integers is even when the number of odd terms is even. Under the product-even condition, this occurs in two ways:

$$\binom{3}{3} = 1 \quad \text{(three evens)},$$

$$\binom{4}{2}\binom{3}{1} = 6 \cdot 3 = 18 \quad \text{(two odds and one even)}.$$

Thus the required probability is

$$\frac{1+18}{31} = \frac{19}{31}.$$

The answer is $\boxed{\text{D}}$.

Q7

Trigonometric symmetry

Answer **D**

Question 7

Which of the following is a line of symmetry of the graph

$$y = \frac{1}{\sin\left(4x + \frac{\pi}{3}\right)}?$$

- **A** $x = \dfrac{13\pi}{2}$
- **B** $x = \dfrac{\pi}{2}$
- **C** $x = \pi$
- **D** $x = \dfrac{13\pi}{24}$
- **E** $y = \dfrac{5\pi}{24}$

WORKED SOLUTION

The reciprocal does not change the vertical symmetry axes of the sine graph. The axes occur when the sine argument is at a maximum or minimum:

$$4x + \frac{\pi}{3} = \frac{\pi}{2} + n\pi.$$

Hence

$$x = \frac{\pi}{24} + \frac{n\pi}{4}.$$

Taking $n = 2$ gives

$$x = \frac{\pi}{24} + \frac{\pi}{2} = \frac{13\pi}{24},$$

which is option D.

Q8

Recurrence and cycles

Answer **A**

Question 8

A sequence is defined by

$$x_{n+1} = \begin{cases} \dfrac{x_n}{2}, & x_n \text{ even}, \\ \dfrac{3x_n+1}{2}, & x_n \text{ odd}. \end{cases}$$

Given $x_1 = 37$, find x_{2026}.

- **A** 1
- **B** 2
- **C** 4
- **D** 5
- **E** 8

WORKED SOLUTION

Iterating only until a cycle appears:

$$37, 56, 28, 14, 7, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1, 2, 1, 2, \ldots$$

Thus $x_{16} = 1$, and from then on the sequence alternates between 1 and 2. Since

$$2026 - 16 = 2010$$

is even, $x_{2026} = x_{16} = 1$. The answer is $\boxed{\text{A}}$.

Q9

Graphs and root counting

Answer C

Question 9

How many solutions does

$$\cos(2x)\ln x = \sin(2x)$$

have in the interval $0 < x < 3\pi$?

A 0

B 3

C 5

D 6

E 7

WORKED SOLUTION

If $\cos 2x = 0$, the right-hand side is ± 1, so no solution is lost by dividing through by $\cos 2x$. The equation becomes

$$\ln x = \tan 2x.$$

On every complete branch

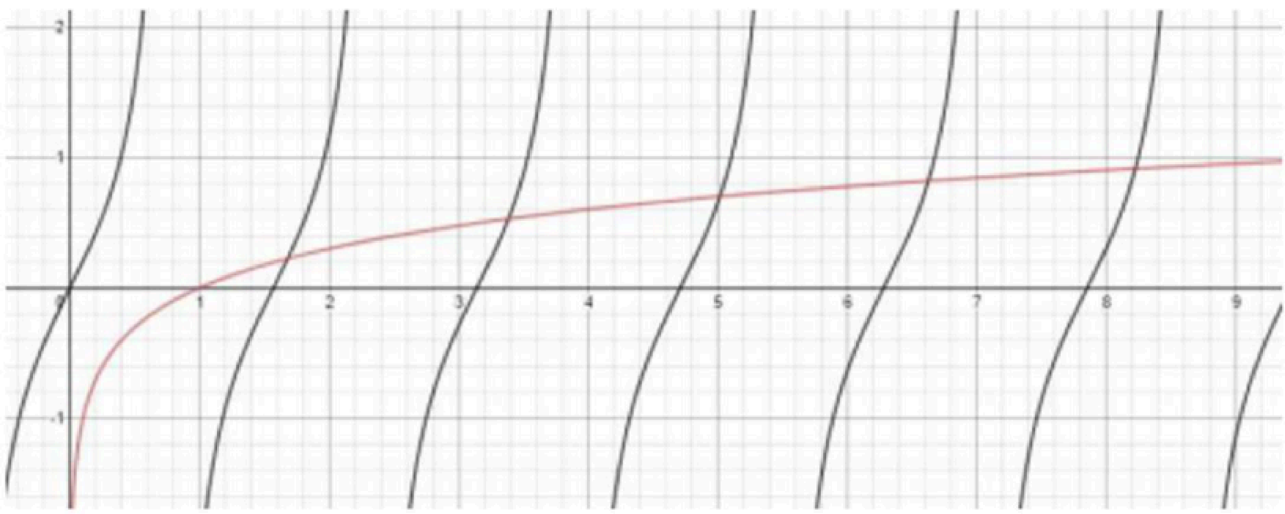
$$\left(\frac{\pi}{4} + \frac{k\pi}{2}, \frac{\pi}{4} + \frac{(k+1)\pi}{2}\right)$$

lying inside $(0, 3\pi)$, the function $\tan 2x - \ln x$ runs from $-\infty$ to $+\infty$. Moreover, for $x > \frac{1}{2}$,

$$\frac{d}{dx}(\tan 2x - \ln x) = 2\sec^2(2x) - \frac{1}{x} > 0,$$

so there is exactly one root on each complete branch.

There are five such complete branches. The initial interval $(0, \pi/4)$ has no root because $\tan 2x > 0$ while $\ln x < 0$, and the final partial branch ends before crossing. Hence there are 5 solutions, so the answer is $\boxed{\text{C}}$.



Graphical comparison of y=tan(2x) and y=ln x on the stated interval.

Functions, Integrals and Recurrences

Integral transformations, Vieta, cubic structure, integer factorisation, limits and definite integrals.

Q10

Integral transformations

Answer D

Question 10

Let f be integrable on $[-4, 4]$ and satisfy

$$f(x) + f(-x) = x^2$$

for all $x \in [-4, 4]$. Given that

$$\int_0^4 f(x)\,dx = 3,$$

evaluate

$$\int_{-2}^{2} \big[f(x-2) - f(2-x) + 1\big]\,dx.$$

A $\frac{46}{3}$

B $\frac{49}{3}$

C $\frac{55}{3}$

D $\frac{58}{3}$

E $\frac{64}{3}$

WORKED SOLUTION

First,

$$\int_{-2}^{2} f(x-2)\,dx = \int_{-4}^{0} f(u)\,du.$$

Using $u = -t$,

$$\int_{-4}^{0} f(u)\,du = \int_0^4 f(-t)\,dt = \int_0^4 \big(t^2 - f(t)\big)dt = \frac{64}{3} - 3 = \frac{55}{3}.$$

Also,

$$\int_{-2}^{2} f(2-x)\,dx = \int_0^4 f(v)\,dv = 3.$$

The constant contributes 4. Therefore

$$I = \frac{55}{3} - 3 + 4 = \frac{58}{3}.$$

The answer is $\boxed{\text{D}}$.

Q11

Vieta and intersections

Answer H

Question 11

The curves

$$y = \pi x(x-1)(x-2)(x-3)(x-4)(x-5)$$

and

$$y = \frac{5}{17}x + \frac{\pi}{3}$$

intersect at six points. Find the sum of the six x-coordinates.

A -15

B $-\frac{44\pi}{51} + 1$

C $-\frac{7}{15}$

D 0

E $\frac{7}{15}$

F $\frac{44\pi}{51} - 1$

G 12

H 15

WORKED SOLUTION

The intersection coordinates are the roots of

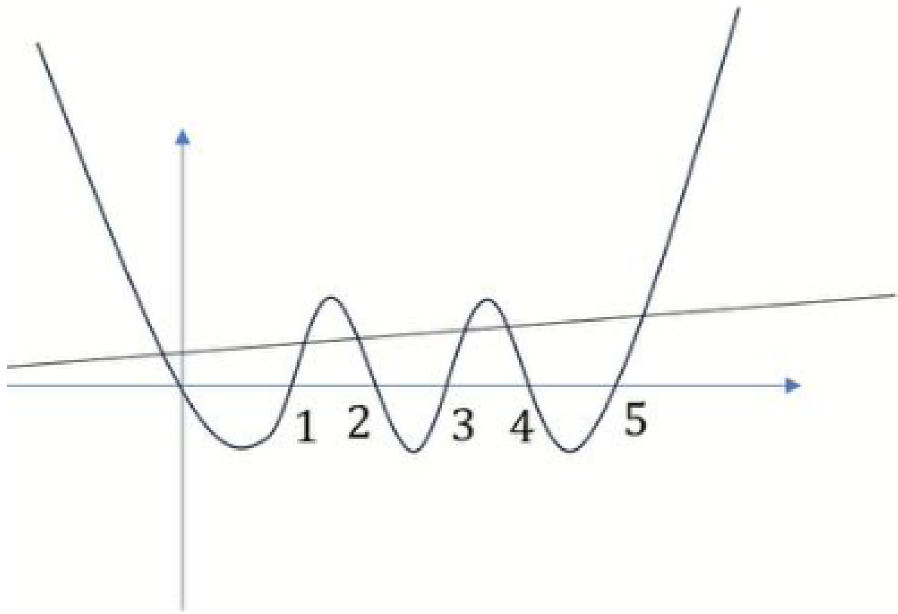
$$\pi x(x-1)(x-2)(x-3)(x-4)(x-5) - \frac{5}{17}x - \frac{\pi}{3} = 0.$$

The product has leading coefficient 1, and its x^5-coefficient is minus the sum $0+1+2+3+4+5=-15$. Therefore the degree-six equation has leading coefficient π and x^5-coefficient -15π. The added linear and constant terms do not affect either coefficient.

By Vieta's formula, the sum of all six roots is

$$-\frac{-15\pi}{\pi} = 15.$$

The answer is $\boxed{\mathrm{H}}$.



A supporting sketch of the degree-six curve and the line; the root sum follows exactly from Vieta’s formula.

Q12

Cubic structure

Answer **E**

Question 12

A monic cubic function f satisfies

$$f(1) = 1, \qquad f(2) = 1, \qquad f(-1) = -5.$$

Find $f(3)$.

- **A** -7
- **B** 0
- **C** 4
- **D** 6
- **E** 7
- **F** 36

WORKED SOLUTION

Since f is monic and $f(1) = f(2) = 1$, the monic cubic $f(x) - 1$ has roots 1 and 2. Hence

$$f(x) - 1 = (x-1)(x-2)(x-r)$$

for some r. Using $f(-1) = -5$,

$$-6 = (-2)(-3)(-1-r) = 6(-1-r),$$

so $r = 0$. Therefore

$$f(x) = x(x-1)(x-2) + 1,$$

and

$$f(3) = 3 \cdot 2 \cdot 1 + 1 = 7.$$

The answer is $\boxed{\text{E}}$.

Q13

Integer factorisation

Answer C

Question 13

Integers a, b, c, each greater than 1, satisfy

$$abc + ab + bc + ca + a + b + c = 104.$$

Find $a + b + c$.

- **A** 10
- **B** 11
- **C** 12
- **D** 13
- **E** 14
- **F** 15

WORKED SOLUTION

Add 1 to both sides:

$$abc + ab + bc + ca + a + b + c + 1 = 105.$$

The left side factors as

$$(a + 1)(b + 1)(c + 1) = 105 = 3 \cdot 5 \cdot 7.$$

Because each of $a + 1, b + 1, c + 1$ is at least 3, the three factors must be $3, 5, 7$ in some order. Thus a, b, c are $2, 4, 6$ in some order, and

$$a + b + c = 12.$$

The answer is $\boxed{\text{C}}$.

Q14

Recurrence limits

Answer C

Question 14

Let

$$u_n = 3u_{n-1} + 2u_{n-2}, \qquad u_1 = 1, \quad u_2 = 5.$$

Find

$$\lim_{k\to\infty} \frac{u_k}{u_{k-1}}.$$

A 2

B $\sqrt{5}$

C $\dfrac{3+\sqrt{17}}{2}$

D $\dfrac{3\pm\sqrt{17}}{2}$

E $\dfrac{3+\sqrt{5}}{2}$

F $1\pm\sqrt{2}$

G $3+\sqrt{2}$

WORKED SOLUTION

The characteristic equation is

$$r^2 - 3r - 2 = 0,$$

with roots

$$r_1 = \frac{3+\sqrt{17}}{2}, \qquad r_2 = \frac{3-\sqrt{17}}{2}.$$

Here $|r_1| > |r_2|$, and the coefficient of the dominant term is non-zero for the given initial values. Therefore

$$\frac{u_k}{u_{k-1}} \longrightarrow r_1 = \frac{3+\sqrt{17}}{2}.$$

The answer is $\boxed{\text{C}}$.

Q15

Definite integral algebra

Answer **D**

Question 15

For every integer $n \geq 0$,

$$\int_n^{n+1} f(x)\,dx = n^2 + n + 1.$$

Evaluate

$$\int_0^3 f(x)\,dx + \int_2^5 f(x)\,dx - 2\int_1^4 f(x)\,dx.$$

- **A** 2
- **B** 4
- **C** 5
- **D** 6
- **E** 8
- **F** 10

WORKED SOLUTION

Let

$$A_n = \int_n^{n+1} f(x)\,dx = n^2 + n + 1.$$

Then

$$A_0, A_1, A_2, A_3, A_4 = 1, 3, 7, 13, 21.$$

The required expression is

$$(A_0 + A_1 + A_2) + (A_2 + A_3 + A_4) - 2(A_1 + A_2 + A_3).$$

Therefore

$$(1 + 3 + 7) + (7 + 13 + 21) - 2(3 + 7 + 13) = 11 + 41 - 46 = 6.$$

The answer is $\boxed{\text{D}}$.

Mixed Problems

Divisibility, transformations, trigonometric equations, stationary points and iterated functions.

Q16

Divisibility in binomial expansions

Answer D

Question 16

In the simplified expansion of

$$(2+3x)^{12},$$

how many terms have a coefficient divisible by 48?

A 0

B 2

C 5

D 9

E 10

F 12

G 13

WORKED SOLUTION

The coefficient of x^k is

$$C_k = \binom{12}{k} 2^{12-k} 3^k.$$

Since $48 = 2^4 \cdot 3$, every $k = 1, 2, \ldots, 8$ works immediately: the factor 3^k supplies a factor 3, and 2^{12-k} supplies at least 2^4.

For $k = 9$,

$$\binom{12}{9} = 220$$

contains an extra factor 2^2, so the total power of 2 is at least 2^5; hence $k = 9$ also works.

For $k = 10, 11, 12$, the total powers of 2 are respectively only $2^3, 2^3, 2^0$, so these fail. The term $k = 0$ has no factor 3.

Thus exactly 9 terms work. The answer is $\boxed{\text{D}}$.

Q17

Graph transformations

Answer **D**

Question 17

The graph of $y = \sqrt{5x - 2}$ undergoes the following transformations in order:

1. translation 4 units left;
2. translation 6 units down;
3. reflection in the x-axis;
4. stretch parallel to the y-axis with scale factor $\frac{1}{3}$;
5. stretch parallel to the x-axis with scale factor 2.

Which equation describes the final graph?

A $y = \sqrt{2 - \dfrac{5x}{18}} - 2$

B $y = \dfrac{\sqrt{-10x - 6} - 4}{3}$

C $y = -\dfrac{\sqrt{\frac{5}{2}x - 22} - 6}{3}$

D $y = -\dfrac{\sqrt{\frac{5}{2}x + 18} - 6}{3}$

E $y = \dfrac{\sqrt{\frac{5}{2}x - 22} - 6}{3}$

F $y = -\dfrac{\sqrt{\frac{5}{2}x + 22} - 6}{3}$

G $y = \dfrac{\sqrt{\frac{5}{2}x + 14}}{3}$

WORKED SOLUTION

Starting with $f(x) = \sqrt{5x - 2}$:

$$\begin{aligned} \text{left } 4 &: \sqrt{5(x+4) - 2} = \sqrt{5x + 18}, \\ \text{down } 6 &: \sqrt{5x + 18} - 6, \\ \text{reflect} &: -\sqrt{5x + 18} + 6, \\ \text{vertical scale } \frac{1}{3} &: \frac{-\sqrt{5x + 18} + 6}{3}. \end{aligned}$$

A horizontal stretch by factor 2 replaces x by $x/2$, giving

$$y = \frac{-\sqrt{\frac{5}{2}x + 18} + 6}{3} = -\frac{\sqrt{\frac{5}{2}x + 18} - 6}{3}.$$

The answer is $\boxed{\mathrm{D}}$.

Q18

Trigonometric equations

Answer **D**

Question 18

How many values of x in

$$-2\pi \le x \le 2\pi$$

satisfy

$$\sin 3x = \cos 2x?$$

A 6	**B** 8
C 9	**D** 10
E 12	**F** 14

WORKED SOLUTION

Write $\cos 2x = \sin\left(\frac{\pi}{2} - 2x\right)$. For $\sin A = \sin B$,

$$A = B + 2k\pi \quad \text{or} \quad A = \pi - B + 2k\pi.$$

The first family gives

$$3x = \frac{\pi}{2} - 2x + 2k\pi \quad \Longrightarrow \quad x = \frac{\pi}{10} + \frac{2k\pi}{5}.$$

The interval condition gives $k = -5, -4, \ldots, 4$, producing 10 values.

The second family gives

$$x = \frac{\pi}{2} + 2k\pi.$$

The two values in the interval, $-\frac{3\pi}{2}$ and $\frac{\pi}{2}$, are already included in the first family. Hence there are 10 distinct solutions. The answer is $\boxed{\text{D}}$.

Q19

Stationary points of trigonometric graphs

Answer C

Question 19

For

$$y = 2\cos 3x + 2\sqrt{3}\sin 3x + 7, \qquad 0 \le x < \pi,$$

find the sum of the x-coordinates of all stationary points.

A π

B $\dfrac{7\pi}{6}$

C $\dfrac{4\pi}{3}$

D $\dfrac{3\pi}{2}$

E $\dfrac{5\pi}{3}$

F 2π

WORKED SOLUTION

Differentiate:

$$\frac{dy}{dx} = -6\sin 3x + 6\sqrt{3}\cos 3x.$$

At a stationary point,

$$\tan 3x = \sqrt{3} \quad \Longrightarrow \quad 3x = \frac{\pi}{3} + k\pi.$$

Hence

$$x = \frac{\pi}{9} + \frac{k\pi}{3}.$$

In $0 \le x < \pi$, this gives

$$\frac{\pi}{9}, \quad \frac{4\pi}{9}, \quad \frac{7\pi}{9}.$$

Their sum is

$$\frac{12\pi}{9} = \frac{4\pi}{3}.$$

The answer is $\boxed{\text{C}}$.

Q20

Iterated absolute values

Answer **B**

Question 20

Let $k > 0$, $f_0(x) = x$, and

$$f_{n+1}(x) = |f_n(x) - k|$$

for non-negative integers n. For $n > 0$, let α and β be the least and greatest real numbers for which $f_n(x) = 0$.

Find

$$\int_\alpha^\beta f_n(x)\,dx$$

in terms of n and k.

- **A** nk^2
- **B** $nk^2 - k^2$
- **C** kn^2
- **D** nk
- **E** $k(n-1)^2$
- **F** $nk^2 - 1$

WORKED SOLUTION

For $n \geq 1$, the zeros of f_n are

$$x = (2j - n + 2)k, \qquad j = 0, 1, \ldots, n-1.$$

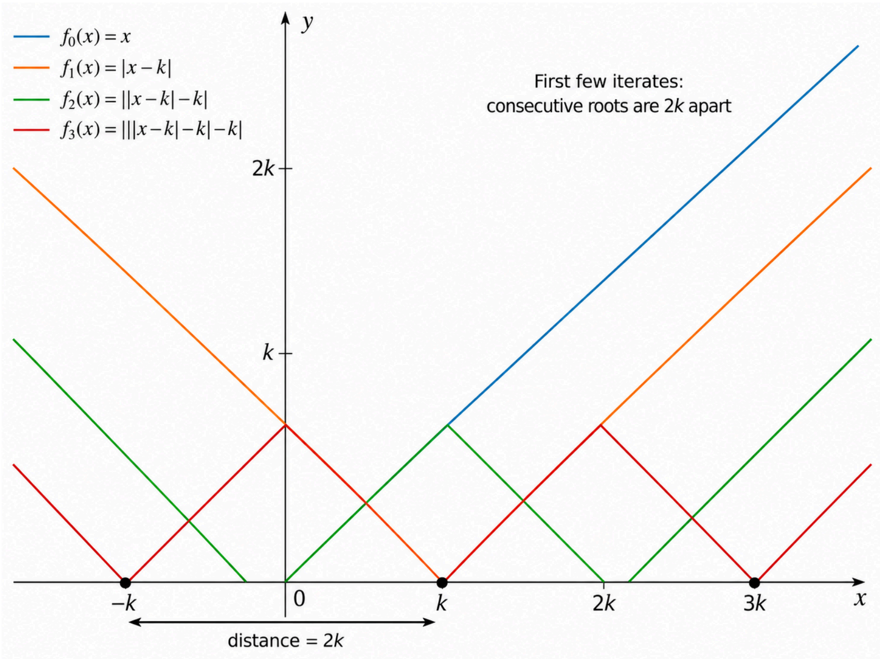
Thus

$$\alpha = (2-n)k, \qquad \beta = nk.$$

Consecutive zeros are $2k$ apart. Between each pair, f_n forms a triangle of base $2k$ and height k. There are $n-1$ such triangles, so

$$\int_{\alpha}^{\beta} f_n(x)\,dx = (n-1)\cdot\frac{1}{2}(2k)(k) = (n-1)k^2 = nk^2 - k^2.$$

The answer is $\boxed{\text{B}}$.



Consecutive zeros are 2k apart; each interval contributes a triangle of area k².